

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Operations Research

Exam: IN0024 / Midterm
Examiner: Prof. Dr. Martin Bichler

Date: Friday 31st May, 2024
Time: 17:00 – 18:00

	P 1	P 2	P 3	P 4
I				
II				

Working instructions

- This exam consists of **12 pages** with a total of **4 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 60 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
 - one **ruler/geo triangle**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Answer the exam questions in English or German.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

Problem 1 Portfolio Optimization (11 credits)

Julia, a financial analyst at *ROI Capital*, wants to create the company's investment plan for 2025 by solving a portfolio optimization problem. *ROI Capital* can invest in 5 assets, $\{A_1, A_2, A_3, A_4, A_5\}$. For each asset A_i with $i \in \{1, \dots, 5\}$, the expected return e_i and the risk measure r_i per Euro invested are known.

Let $x_i \in \mathbb{R}_{\geq 0}$ with $i \in \{1, \dots, 5\}$ represent the amount of money (in Euros) invested in asset A_i . Julia must consider multiple constraints in her portfolio optimization problem. Help Julia decide the amount of money that the company should invest in each asset.

Formulate the following constraints using only linear (in)equalities.



a) The amount of money invested in A_1 and A_3 should be equal to 30% of the amount of money invested in A_2 and A_5 .

$$x_1 + x_3 = 0.3(x_2 + x_5)$$

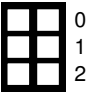


b) The absolute difference between the amount of money invested in A_1 and A_2 should not exceed 15,000 Euros.

$$x_1 - x_2 \leq 15,000 \text{ and } x_2 - x_1 \leq 15,000$$

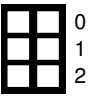
c) Julia wants to maximize the expected return. Formulate a suitable linear objective.

$$\max \sum_{i \in \{1, \dots, 5\}} e_i x_i$$



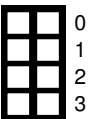
d) Julia wants to minimize the risk. Formulate a suitable linear objective

$$\min \sum_{i \in \{1, \dots, 5\}} r_i x_i$$



e) Julia has access to a solver that supports only single-objective linear programs. Formulate a linear objective that considers the expected return maximization and risk minimization objectives, assuming that for Julia maximizing the expected return is three times more important than minimizing the risk.

$$\max 3E - R, \text{ where } E = \sum_{i \in \{1, \dots, 5\}} e_i x_i \text{ and } R = \sum_{i \in \{1, \dots, 5\}} r_i x_i$$

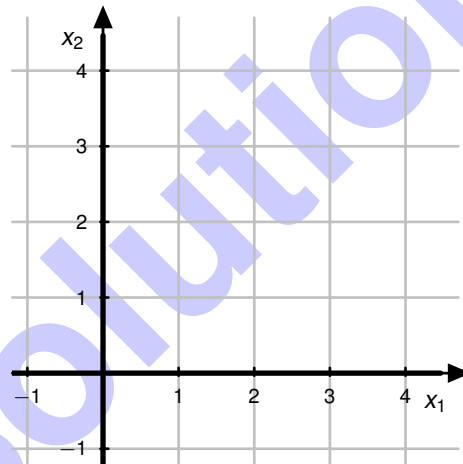
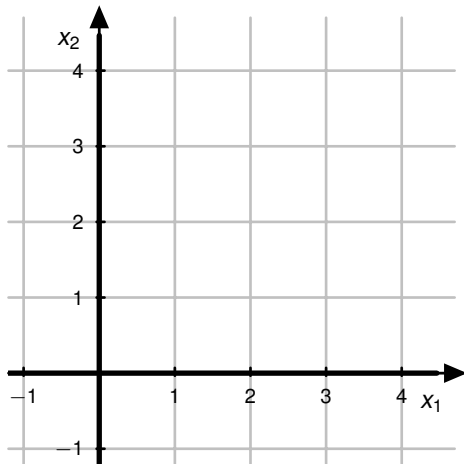


Problem 2 LP Mapping (14 credits)

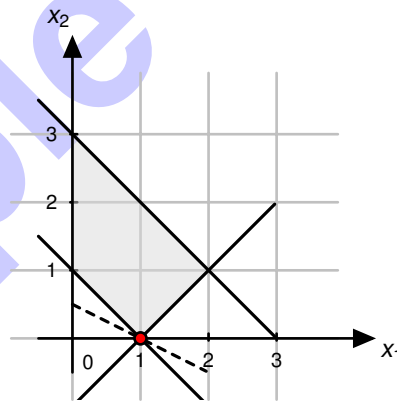
a) Solve the LP given in (P_1) graphically. Mark the feasible region and state the optimal vertex as well as the optimal objective value.

$$\begin{aligned}
 (P_1) \quad & \text{Min} \quad x_1 + 2x_2 \\
 & \text{s.t.} \quad \begin{aligned}
 & x_1 + x_2 \geq 1 \\
 & x_1 + x_2 \leq 3 \\
 & x_1 - x_2 \leq 1 \\
 & x \in \mathbb{R}_{\geq 0}^2
 \end{aligned}
 \end{aligned}$$

Note: Use the provided coordinate systems below.

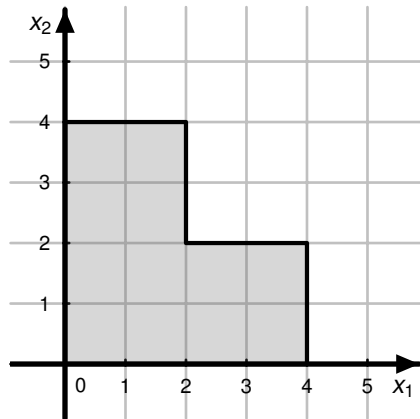


The optimal vertex is $x^* = (1, 0)$ and the optimal objective value is $c^T x^* = 1$.



	0
	1
	2

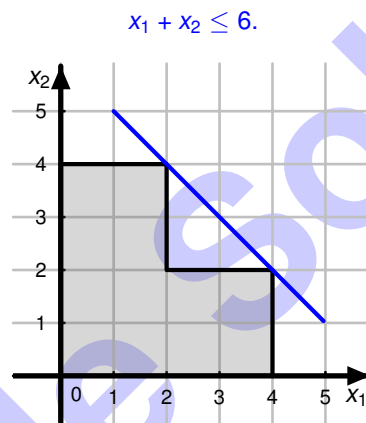
b) Consider the feasible region marked in grey. Suppose you want to optimize a linear objective over this non-convex set. Your colleague has already modelled the feasible region with constraints (P_2) .



$$(P_2) \quad \begin{aligned} x_1 &\leq 4 \\ x_2 &\leq 4 \\ \min\{x_1, x_2\} &\leq 2 \\ x &\in \mathbb{R}_{\geq 0}^2 \end{aligned}$$

However, the constraint $\min\{x_1, x_2\} \leq 2$ is not linear. Model this constraint with linear (in)equalities such that all optimal solutions of the original problem are optimal in the new LP under all possible linear objectives. Additionally, draw the corresponding linear (in)equalities.

A suitable constraint is:



c) Give an example objective and solution, where an optimal solution of your new LP is not feasible for the original problem.

	0
	1
	2

For $\max x_1 + x_2$, the solution $x^* = (x_1^*, x_2^*) = (3, 3)$ is optimal and feasible for the new LP but infeasible for the original one.

d) Can the simplex algorithm on the new problem return a point which is not feasible for the original one? Justify.

	0
	1
	2
	3

The simplex algorithm always returns a vertex. The vertices of the new LP are feasible for the original problem. Thus, the simplex algorithm never returns an infeasible point.

Problem 3 Simplex - First Fit (20 credits)

In the simplex algorithm, it is conventionally suggested to choose a column with the smallest negative coefficient in the objective row as the pivot column. This corresponds to progressing within the solution space in the direction of the *steepest ascent/descent* of the objective function value. However, it is also possible to choose any column with a negative coefficient. In each step, this also leads to an improvement in the solution, but may require more iterations. In the *first fit method*, the objective row is considered from left to right and the first column with a negative coefficient is selected.

Note: The objective row is assumed to be ordered s.t. x_i is left to x_j , whenever $i < j$, e.g., x_1 is left to x_2 .

a) Transform the LP (P_S) into canonical form.

$$\begin{aligned} (P_S) \quad & \text{Max} && x_1 + 2x_2 \\ & \text{s.t.} && \\ & && x_1 + x_2 \leq 100 \\ & && 6x_1 + 9x_2 \leq 720 \\ & && x_2 \leq 60 \\ & && x \in \mathbb{R}_{\geq 0}^2 \end{aligned}$$

$$\begin{aligned} \text{Max} &&& x_1 + 2x_2 \\ \text{s.t.} &&& \\ &&& x_1 + x_2 + x_3 = 100 \\ &&& 6x_1 + 9x_2 + x_4 = 720 \\ &&& x_2 + x_5 = 60 \\ &&& x \in \mathbb{R}_{\geq 0}^5 \end{aligned}$$

b) Solve the LP (P_S) with the simplex algorithm in tableau form using the steepest ascent method. In each iteration state/highlight the pivot element and the corresponding incoming and outgoing basic variable. State the optimal solution and the optimal objective.

Note: Use the tableaus provided on the next page.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	-1	-2	0	0	0	0
x_3	1	1	1	0	0	100
x_4	6	9	0	1	0	720
x_5	0	1	0	0	1	60

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	-1	0	0	0	2	120
x_3	1	0	1	0	-1	40
x_4	6	0	0	1	-9	180
x_2	0	1	0	0	1	60

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	0	0	1/6	1/2	150
x_3	0	0	1	-1/6	1/2	10
x_1	1	0	0	1/6	-3/2	30
x_2	0	1	0	0	1	60

Stopping criterion fulfilled (only positive entries in the objective row), thus $x = (30, 60)$ and $z_{\max} = 150$ are the optimal solution.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

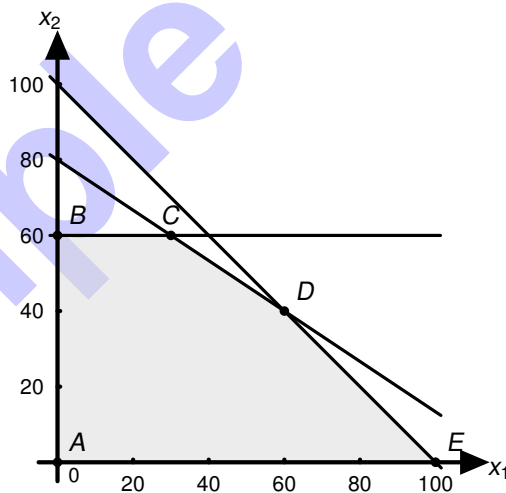
Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

c) List the sequence of points visited (in $\mathbb{R}^2$) in the steepest ascent method and in the first fit method on the path to the optimum. Additionally, justify the order in the sequences.

	0
	1
	2
	3
	4
	5



In both cases the paths start in A and end in C. A corresponds to the initial basis and C is optimal.
The objective is $\max c_1 x_1 + c_2 x_2$ with $c_1 = 1 < 2 = c_2$.
Steepest ascent: A,B,C
This can be seen from the simplex tableaus computed in part b) or graphically by first increasing x_2 and then x_1 , since $c_2 > c_1$.
First-fit: A,E,D,C
We first increase x_1 and then increase x_2

Problem 4 Sensitivity Analysis (15 credits)

A textile factory produces two types of coats: one for women and one for men. The profit of producing one coat is 300€ for the women's type and 200€ for the men's type. The factory needs leather, machine time, and manual labor for production. In the table below, you can see the required resources and the profits for each product.

Product	Leather Supplies	Machine Time	Manual Time	Profit
Women's coat	10dm ²	20h	40h	300€
Men's coat	15dm ²	55h	10h	200€

The factory has a limited amount of each resource: 2,000dm² of leather, 7,000h of machine time and 4,000h of manual working time are available per month.

An employee has already calculated an optimal simplex tableau. The slack variables $s_1, s_2, s_3 \in \mathbb{R}_{\geq 0}$ correspond to the leather, machine time and manual working time resources, respectively.

Base	Coefficients					Res.
	x_1	x_2	s_1	s_2	s_3	
z	0	0	10	0	5	40,000
x_2	0	1	2/25	0	-1/50	80
s_2	0	0	-4	1	1/2	1000
x_1	1	0	-1/50	0	3/100	80

Hint: You can use the provided tableaus for this task. Additional tableaus can be found on the last pages.

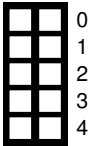
a) Determine the shadow prices for all resources. Find the limits for the availability of the resources under which the shadow prices do not change.

We determine the shadow prices from the optimal tableau. These are 10€ for leather, 0€ for machine time, and 5€ for manual time. To find the limits for the resources under which the shadow prices do not change, we need to multiply the corresponding slack variable's column by Δ and add it to the right side of the simplex tableau. The shadow prices would not change as long as the current basis stays feasible. The current basis stays feasible as long as the values on the right side of the simplex tableau stay non-negative. For the leather resource, this gives inequalities: $80 + \frac{2}{25}\Delta_1 \geq 0$, $1000 - 4\Delta_1 \geq 0$ and $80 - \Delta_1/50 \geq 0$. The limits for the change in the availability of the leather resource are $\Delta_1 \in [-1000, 250]$. For the machine time resource, we only have $1000 + \Delta_2 \geq 0$. The limits for the change in the availability of the machine time resource are $\Delta_2 \in [-1000, \infty)$. For the working time resource we have $80 - \Delta_3/50 \geq 0$, $1000 + \Delta_3/2 \geq 0$ and $80 + \frac{3}{100}\Delta_3 \geq 0$. The limits are $\Delta_3 \in [-2000, 4000]$.

Base	Coefficients					Res.

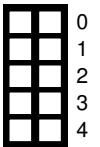
Base	Coefficients					Res.

- b) Calculate the minimum profit at which the factory would agree to produce a women's boot that requires $8dm^2$ leather, $10h$ machine time, and $50h$ working time.



Producing a boot reduces the resources available for coats. From our previous answer, we know that the current shadow prices do not change if we reduce the resources by the amount that we need to produce one women's boot. So we can use the current optimal shadow prices. To do this, we evaluate the resource requirements of the women's boot with their shadow prices and get $8 \cdot 10 + 10 \cdot 0 + 50 \cdot 5 = 330€$. This means that the minimum profit for the women's boot should be 330€ before it should be produced.

- c) Calculate within which profit interval for the women's coat the current optimal solution would not change.



We want to know how the objective function coefficient c_1 can change. To do this, we change it to $c_1 = 300 + \Delta$. This gives us the following final tableau:

Base	Coefficients					Res.
	x_1	x_2	s_1	s_2	s_3	
z	$-\Delta$	0	10	0	5	40,000
x_2	0	1	$2/25$	0	$-1/50$	80
s_2	0	0	-4	1	$1/2$	1000
x_1	1	0	$-1/50$	0	$3/100$	80

We now carry out necessary calculations to have 0 coefficient in the column of z for the basic variable x_1 . We get the following tableau:

Base	Coefficients					Res.
	x_1	x_2	s_1	s_2	s_3	
z	0	0	$10 - \Delta/50$	0	$5 + 3\Delta/100$	40,000
x_2	0	1	$2/25$	0	$-1/50$	80
s_2	0	0	-4	1	$1/2$	1000
x_1	1	0	$-1/50$	0	$3/100$	80

All coefficients in the objective function row must be non-negative, which means $10 - \frac{1}{50}\Delta \geq 0$ and $5 + \frac{3}{100}\Delta \geq 0$, so $\Delta \in [-\frac{500}{3}, 500]$. To keep the women's coat in the basis, the following need to be satisfied: $300 - \frac{500}{3} \leq c_1 \leq 300 + 500 = 800$.

Base	Coefficients					Res.

Base	Coefficients					Res.

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

Sample

Sample Solution

Sample Solution